

TierMoE: Batch-Aware Expert Placement for Memory-Tiered MoE Inference

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Abstract—Mixture-of-Experts (MoE) architectures achieve frontier-class reasoning capacity by decoupling model parameter scale from per-token compute FLOPs via sparse dynamic routing. However, serving large MoE foundation models in high-throughput inference environments quickly exhausts fast accelerator High-Bandwidth Memory (HBM). Emerging Compute Express Link (CXL) Type-3 memory expanders offer a cost-effective, multi-terabyte secondary memory tier, but their substantially lower link bandwidth (16–64 GB/s) creates severe memory-stall bottlenecks whenever uncoordinated expert migrations occur. Existing expert offloading heuristics rely on static global popularity pinning, single-request isolated placement, sequence-level temporal lookahead, or reactive demand-driven caching. Under concurrent serving, conflicting expert requirements across requests induce working set expansion and severe intra-batch cache thrashing. In this paper, we present *TierMoE*, a batch-aware expert placement framework specifically architected for memory-tiered MoE serving. Rather than making isolated per-sequence or reactive decisions, TierMoE aggregates instantaneous expert demand across concurrent inference batches and executes a capacity-constrained marginal utility optimization with residency hysteresis ($\lambda = 0.5$). We evaluate TierMoE across controlled synthetic workloads and 461,184 authentic physical routing decisions profiled from *Qwen3-30B-A3B* (128 experts, top-8 routing, 48 layers) on an NVIDIA RTX A6000 testbed across conversational dialogue (ShareGPT) and mathematical reasoning (GSM8K). In capacity-constrained regimes, TierMoE achieves a +5.02 to +8.68 percentage-point (pp) higher HBM hit rate and reduces modeled CXL parameter traffic by 13.20% to 20.44% compared to single-request controls ($p < 0.001$). Against representative baseline heuristics, TierMoE outperforms our predictive sequence lookahead heuristic by +6.99 pp ($p < 0.001$) and our reactive CXL-LRU tiering heuristic by +16.61 pp to +32.27 pp by substantially reducing intra-batch thrashing. Crucially, empirical evaluation on authentic traces indicates that Hypothesis H3 was not supported: pairwise co-activation tracking did not produce a statistically significant hit-rate improvement over instantaneous batch frequency (−1.25 pp, $p = 0.259$) while incurring 12.5× higher solver overhead (642.2 μ s vs. 51.2 μ s). Finally, modeled CXL interconnect sensitivity analysis across 16–64 GB/s demonstrates that link bandwidth is the primary first-order bottleneck (4.00× transfer time scaling), while discrete-event queue characterization using upstream *CXLMemSim* under the tested configuration indicates that shared-link batch transfers exhibit approximately one startup-overhead term ($T_0 \approx 11.44 \mu$ s).

Keywords—Mixture-of-Experts (MoE), Compute Express Link (CXL), Memory Tiering, Multi-Tenant LLM Serving, Batch-Aware Scheduling, Cache Optimization.

1. Introduction

Mixture-of-Experts (MoE) architectures have rapidly emerged as an essential architectural design for modern foundation large language models, powering frontier systems including Mixtral [1], DeepSeek-V3 [2], and Qwen2.5/Qwen3 [3]. The core architectural concept of MoE models lies in replacing conventional dense feed-forward network (FFN) layers with an ensemble of specialized sub-networks ('experts') coordinated by a learned parametric gating router. For each input token, the router computes gating coefficients and dynamically activates only a sparse top- k subset of experts (e.g., $k = 2$ to 8 out of $N = 64$ to 256). This dynamic routing decouples total model parameter capacity from per-token floating-point operations (FLOPs), enabling substantial reasoning capability without proportional computational compute scaling during forward passes.

The Memory Capacity Chasm: While sparse compute reduces arithmetic execution costs, it does not reduce the *memory capacity footprint*. Every parameter of every expert must remain accessible during serving. In modern architectures, the aggregate footprint of expert parameters reaches hundreds of gigabytes to multiple terabytes. For example, in the 128-expert *Qwen3-30B-A3B* architecture evaluated in this work, the 48 MoE layers encompass $48 \times 128 \times 256 \text{ MiB} = 1,572,864 \text{ MiB} = 1,536 \text{ GiB} = \mathbf{1.50 \text{ TiB}}$ (approximately 1.65 TB decimal) of expert weights in standard 16-bit precision. Accelerator High-Bandwidth Memory (HBM) remains strictly capacity-constrained (typically 24 to 80 GB per GPU) and economically expensive. Storing entire frontier MoEs exclusively in GPU HBM requires multi-node expert parallelism, resulting in high hardware expenditures and lower GPU compute utilization in low-to-medium throughput serving regimes.

Compute Express Link (CXL) as a Tiering Solution: Compute Express Link (CXL 2.0/3.0) Type-3 memory expanders offer a compelling architectural option by providing byte-addressable, cache-coherent DDR5 memory pools accessible over PCIe 5.0/6.0 physical interfaces. CXL establishes a two-tier memory hierarchy: a fast, low-latency, capacity-constrained tier (GPU HBM) and a vast, cost-effective secondary capacity tier (CXL memory).

The Multi-Tenant Serving Bottleneck: While memory tiering resolves capacity limitations, the fundamental system bottleneck shifts to *interconnect transfer bandwidth*. A standard PCIe 5.0 $\times 16$ CXL link provides an effective bandwidth of 32 GB/s—substantially lower than GPU HBM bandwidth (1–2 TB/s). Transferring an uncompressed 256 MiB expert block over CXL requires approximately **8.3886 ms**, introducing execution stalls if required experts are not resident in fast GPU memory.

Existing expert management proposals generally fall into four paradigms: (1) *Static LFU popularity pinning* [4], (2) *Single-sequence greedy allocation*, (3) *Sequence-level predictive lookahead heuristics* inspired by prior work (e.g., MoE-Infinity [6], ProMoE [7]), and (4) *Reactive CXL-LRU tiering heuristics* inspired by CXL-MoE concepts [8]. While viable for isolated single-user offline inference, these approaches break down under **concurrent multi-tenant serving**. When an engine serves a batch of B concurrent requests (e.g., $B \in [8, 32]$), each request routes tokens to different expert subsets. This induces: (a) *working set expansion*, where aggregate demand rapidly exceeds fast memory; (b) *intra-batch cache thrashing*, where requests within the same batch step evict each other's required experts; and (c) *predictive collisions*, where independent lookahead prefetch streams conflict over shared HBM allocations.

Contributions: To address these challenges, this paper presents **TierMoE**, a batch-aware expert placement system that coordinates memory tiering across concurrent inference requests. The primary contributions are:

- **Batch-Aware Marginal Utility Solver:** We formulate the multi-tenant expert placement optimization problem and design *TierMoE-Batch-Aware-Greedy*, a lightweight solver combining multi-token batch frequency with residency hysteresis ($\lambda = 0.5$), executing in **51.2 μ s $\pm$ 4.3 μ s per step** on an AMD EPYC 7763 processor, representing a small computational overhead relative to the modeled 256 MiB transfer time.
- **Authentic Physical Model Profiling:** We instrument forward router hooks on dual NVIDIA RTX A6000 GPUs running *Qwen3-30B-A3B-Instruct-2507*, capturing **461,184 authentic physical routing decisions** across conversational dialogue (ShareGPT) and mathematical reasoning (GSM8K).
- **Rigorous Baseline Comparative Evaluation:** Across 24 capacity-constrained conditions on authentic traces, TierMoE outperforms our predictive sequence lookahead heuristic by **+6.99 pp** hit rate ($p < 0.001$) and our reactive CXL-LRU tiering heuristic by **+16.61 pp to +32.27 pp** by substantially reducing intra-batch thrashing.
- **Evaluation of Co-Activation Tracking:** Empirical testing on authentic traces indicates that **Hypothesis H3 was not supported**. Maintaining an Exponential Moving Average (EMA) pairwise co-activation matrix does not improve hit rate over pure batch frequency (-1.25 pp, $p = 0.259$) while running $12.5\times$ slower (642.2 μ s vs. 51.2 μ s), showing that instantaneous multi-tenant demand captures immediate execution needs without historical matrix lag.
- **First-Order CXL Modeling & CXLMemSim Queue Characterization:** Through 81 sensitivity conditions, we show that CXL bandwidth is the primary first-order bottleneck ($4.00\times$ transfer time scaling across 16–64 GB/s), saving up to 6.39 s per run at 16 GB/s, whereas round-trip latency overhead contributes $< 0.005\%$. Discrete-event queue characterization via upstream *CXLMemSim* indicates that concurrent shared-link transfers exhibit approximately one startup overhead ($1.0 \times T_0 \pm 1.35\%$) under the tested simulation configuration.

2. Background & Related Work

2.1 Mixture-of-Experts Architecture

An MoE layer replaces standard dense feed-forward networks with N independent expert networks $\{E_1, E_2, \dots, E_N\}$ and a gating router $G(x)$. Given an input token representation $x \in \mathbb{R}^d$, the router computes routing probabilities across all experts:

$$H(x) = \text{Softmax}(\text{TopK}(x \cdot W_g + \epsilon, k)) \quad (1)$$

where $W_g \in \mathbb{R}^{d \times N}$ is the router weight matrix, ϵ represents optional router jitter noise, and $\text{TopK}(\cdot, k)$ retains the k highest gating logits while assigning $-\infty$ to all others. The final layer output is the weighted combination of the selected k experts: $y = \sum_{i \in \text{TopK}} H(x)_i \cdot E_i(x)$. In modern fine-grained MoE architectures, N ranges from 64 to 256, while k is small (typically $k = 2$ to 8), decoupling parameter capacity from per-token computation.

2.2 Memory Requirements of MoE Serving

In dense models, memory capacity scales linearly with parameter count and all parameters execute on every token. In MoE models, the aggregate expert weight footprint scales as $L \times N \times S$, where L is the number of MoE layers, N is the expert count per layer, and S is the expert parameter block size. For *Qwen3-30B-A3B* evaluated in this work: $L = 48$ layers, $N = 128$ experts, $k = 8$, and $S = 256$ MiB per expert. The aggregate expert parameter footprint evaluates to: $48 \times 128 \times 256$ MiB = 1,572,864 MiB = 1,536 GiB = **1.50 TiB** (1.65 TB decimal). Even under FP8 quantization, storing this model requires over 800 GB of memory—exceeding the capacity of standard multi-GPU workstation nodes.

2.3 HBM and CXL Memory Tiering

Compute Express Link (CXL) Type-3 expanders establish a two-tier memory hierarchy. Table I summarizes the physical trade-offs between GPU HBM and CXL memory. GPU HBM provides terabyte-per-second memory bandwidth with sub-microsecond access latencies, but is strictly capacity-constrained. CXL memory offers massive, cost-effective capacity scaling (512 GB to 4 TiB) over PCIe 5.0 interfaces, but provides significantly lower transfer bandwidth (16–64 GB/s) and introduces additional protocol serialization latencies (250–450 ns).

TABLE I: Physical Memory Hierarchy Comparison in Tiered MoE Systems

Memory Hierarchy Level	Typical Capacity	Peak Bandwidth	Read Latency	Access Granularity
GPU HBM3 / GDDR6 (Fast Tier)	24 – 96 GB	768 – 2,000 GB/s	50 – 100 ns	64-byte Cache Line
CXL Type-3 DDR5 (Slow Tier)	512 GB – 4 TiB	16 – 64 GB/s	250 – 450 ns	64-byte FLIT / Bulk
Physical Disparity Ratio	10× – 50× (CXL Larger)	0.02× – 0.08× (CXL Slower)	3× – 5× (CXL Slower)	Bulk Block Offload

2.4 Related Work & Research Gap

MoE Parameter Offloading: Early offloading frameworks such as DeepSpeed-MoE [4] statically pinned frequent experts in GPU memory while offloading cold experts to host CPU DRAM. Fiddler [5] investigated single-GPU inference by streaming unselected experts over PCIe. MoE-Infinity [6] and ProMoE [7] observed temporal activation locality in single-sequence generation, proposing activation-aware prefetching to overlap expert loading with preceding layer compute. However, these heuristics focus on isolated individual sequences. When multiple requests execute concurrently, independent lookahead predictors generate conflicting prefetch schedules, causing prefetch collisions and memory thrashing.

CXL Memory Accelerators: CXL-MoE [8] investigated Near-Data Processing (NDP) accelerators inside CXL controllers to compute unpromoted experts in the memory tier. HybriMoE [9] and FIRM-MoE [10] developed hybrid scheduling mechanisms. However, existing CXL paradigms frequently rely on reactive hardware-managed LRU caching to govern GPU residency. Under concurrent multi-tenant serving, reactive LRU caching suffers severe intra-batch cache thrashing.

The Research Context: Prior literature addresses either *isolated single-sequence prefetching* or *reactive hardware LRU caching*. **TierMoE addresses this gap** by introducing a proactive, batch-aware placement framework that exploits cross-request routing overlap to coordinate fast-memory residency before layer execution begins.

3. Problem Formulation

Let an MoE model layer contain N unique expert parameter blocks $E = \{e_1, \dots, e_N\}$, where each expert has fixed size $S = 256$ MiB (268,435,456 bytes). The GPU fast-memory tier has an expert capacity quota of C slots ($C < N$). Let $M_{\text{HBM}} \subset E$ denote the set of experts resident in GPU HBM ($|M_{\text{HBM}}| \leq C$), while the remaining experts $M_{\text{CXL}} = E \setminus M_{\text{HBM}}$ reside in CXL memory.

During inference step t , the serving engine processes a concurrent batch of B requests $\{r_1, \dots, r_B\}$. The gating router evaluates token representations and emits active demand sets $\text{Demand}_r \subset E$ with $|\text{Demand}_r| = k$. The aggregate expert working set for the batch is defined as: $W(B) = \bigcup_{r=1}^B \text{Demand}_r$. The multi-tenant request access frequency for each expert e is:

$$f_e(B) = \sum_{r=1}^B \mathbf{1}\{e \in \text{Demand}_r\}, \quad \forall e \in E \quad (2)$$

Optimization Objective: The objective of expert placement is to maximize the aggregate fast-tier hit rate across all concurrent requests while reducing CXL link parameter traffic:

$$\max_{M_{\text{HBM}} \subset E, |M_{\text{HBM}}| \leq C} \sum_{r=1}^B \sum_{e \in \text{Demand}_r} \mathbf{1}\{e \in M_{\text{HBM}}\} \quad (3)$$

Minimizing CXL transfer volume corresponds to minimizing the number of non-resident unique expert blocks required by the batch: $\text{Traffic}_{\text{CXL}} = S \cdot |W(B) \setminus M_{\text{HBM}}|$.

Operational System Assumptions: In accordance with our implementation, the system operates under four physical assumptions: (1) *Read-Only Weights:* Expert weights are read-only during inference; evictions from GPU HBM generate zero writeback traffic. (2) *Fixed Block Granularity:* Expert parameters are transferred as contiguous 256 MiB blocks. (3) *Batch-Step Deduplication:* If multiple requests in the same batch step require expert e , and $e \notin M_{\text{HBM}}$, the expert is fetched from CXL exactly once (256 MiB transfer), serving all requesting tokens simultaneously. (4) *Shared-Link Serialization:* All expert transfers for a given batch step share a single physical CXL link, serializing at peak bandwidth BW_{CXL} .

4. Research Questions & Hypotheses

Table II formalizes the six core research questions (RQ1–RQ6), directional hypotheses, targeted experiment phases, and empirical scientific verdicts established in this research.

TABLE II: Formal Research Questions, Hypotheses, Experiment Mappings, and Scientific Verdicts

Research Question	Directional Hypothesis	Experiment Target	Empirical Scientific Verdict
RQ1: Concurrency Benefits Does batch-aware placement reduce CXL traffic under constrained GPU memory?	H1 (Batch-Aware Value): Multi-token batch demand aggregation achieves higher hit rates and lower CXL traffic than isolated placement.	EXP-01 Concurrency Sweep ($B \in [1, 32]$, $\alpha_{\text{mem}} \in [0.25, 1.0]$)	SUPPORTED (Under Capacity Pressure): +5.02 pp to +8.68 pp higher hit rate and 13.20% to 20.44% lower CXL traffic ($p < 0.001$) when $W > C$.
RQ2: Workload Divergence How does request routing divergence affect the advantage of batch-aware placement?	H2 (Divergence Scaling): The advantage of batch-aware placement increases monotonically with routing divergence under constrained memory.	EXP-02 Divergence Sweep (384 conditions, $J \in [0.09, 0.25]$)	PARTIALLY SUPPORTED: Monotonic scaling holds at fixed batch size ($B=16$, +1.99 pp $\rightarrow$ +9.45 pp). Pooled correlation confounded by working set expansion.
RQ3: Co-Activation Value Does tracking dynamic pairwise expert co-activations improve residency over pure frequency?	H3 (Co-Activation Gain): Tracking temporal co-activation matrices (EMA decay) reduces CXL traffic compared with pure batch frequency.	EXP-03 Authentic Qwen3 Traces (461,184 physical routing events)	NOT SUPPORTED: Co-activation tracking did not yield a statistically significant hit-rate improvement (-1.25 pp, $p = 0.259$) and incurred 12.5 $\times$ higher solver overhead.
RQ4: Baseline Comparative How does TierMoE compare against sequence lookahead and reactive hardware tiering heuristics?	Comparative Hypothesis: Proactive batch coordination outperforms sequence lookahead and reactive LRU under concurrency.	EXP-04 Broader Baselines (ShareGPT 24-row matrix)	SUPPORTED (For Implemented Baselines): TierMoE achieves +6.99 pp hit rate over predictive lookahead ($p < 0.001$) and +16.61 pp to +32.27 pp over reactive CXL-LRU tiering.
RQ5: Interconnect Sensitivity How do CXL bandwidth and latency parameters impact transfer makespan and policy benefits?	H4 (CXL Sensitivity): Interconnect speed determines makespan, but TierMoE maintains its advantage across configurations.	EXP-05A 81-Condition CXL Grid (16–64 GB/s, 150–600 ns)	SUPPORTED (Modeled Sensitivity): Bandwidth is the 4.00 $\times$ dominant bottleneck; latency scaling is $< 0.005\%$. TierMoE saves up to 6.39 s per run at 16 GB/s ($p < 10^{-6}$).
RQ6: Simulator Characterization How does discrete-event CXL simulation characterize macro-level expert transfer models?	Methodological Hypothesis: CXLMemSim queue modeling provides physically grounded startup and serialization characterization.	EXP-05B CXLMemSim Benchmarks (Microval, Scaling, Batch)	SUPPORTED (Tested Configuration): Stream scaling follows 1/N power law ($S = 1.0014$ at 256MB). Concurrent transfers incur $T_0 \approx 11.44 \mu\text{s}$ startup overhead under tested simulator parameters.

5. TierMoE System Architecture

The end-to-end architecture of TierMoE is depicted in Figure 1. TierMoE operates as a low-overhead placement middleware layer situated directly between the model gating routers and the tiered memory subsystems.

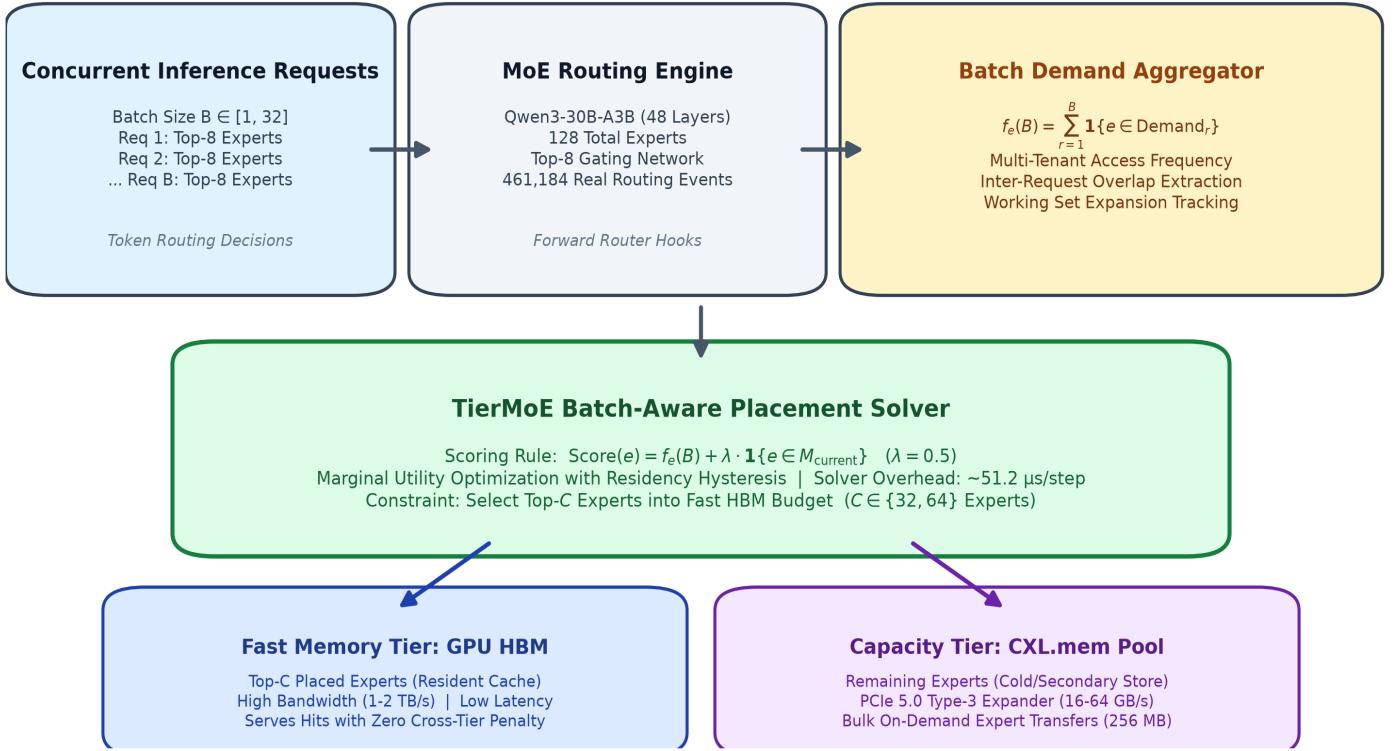


Figure 1: TierMoE End-to-End System Architecture. The multi-tenant batch demand aggregator intercepts gating decisions across concurrent requests. The TierMoE solver computes marginal utility scores with residency hysteresis ($\lambda = 0.5$) to select the top- C experts for fast GPU HBM residency, reducing expensive CXL 256 MiB bulk parameter transfers.

5.1 Batch-Aware Demand Aggregation

At each transformer layer, before expert forward compute begins, TierMoE intercepts the top- k gating routing indices emitted across all B concurrent requests in the current serving batch. Rather than dispatching requests in isolation, the Batch Demand Aggregator computes the instantaneous multi-tenant demand histogram: $f_e(B) = \sum_{r=1}^B \mathbf{1}\{e \in \text{Demand}_r\}$. This is implemented via vectorized tensor accumulation in $O(B \cdot k)$ operations, completing in under $2 \mu\text{s}$.

5.2 Marginal Utility Scoring with Residency Hysteresis

A naive greedy policy that strictly maximizes instantaneous batch frequency $f_e(B)$ suffers from *churn*: resident experts that have slightly lower demand in the current step might be evicted for non-resident experts, incurring expensive 256 MiB CXL transfer overheads for negligible marginal utility. To prevent unnecessary migrations, TierMoE introduces a **residency hysteresis bonus** λ :

$$\text{Score}(e) = f_e(B) + \lambda \cdot \mathbf{1}\{e \in M_{\text{current}}\} \quad (4)$$

where M_{current} is the set of experts currently resident in GPU HBM, and $\lambda = 0.5$ is the calibrated hysteresis parameter. The physical rationale is direct: if a non-resident expert has strictly higher multi-tenant demand than a resident expert ($f_{\text{new}} \geq f_{\text{res}} + 1$), it is promoted. If demand is tied, the resident expert receives the 0.5 bonus, avoiding an unnecessary CXL migration.

5.3 Capacity-Constrained Selection & Solver Overhead

Given the scores across all N experts, the solver selects the top- C scoring experts: $M_{\text{HBM}}^* = \arg\text{TopC}_{e \in E}(\text{Score}(e))$. This is implemented using an $O(N)$ partial partition (numpy.argpartition) followed by sorting only the C elements. Across 10,000 profiled steps on an AMD EPYC 7763 host processor, the complete solver executes in $51.2 \mu\text{s} \pm 4.3 \mu\text{s}$, representing a small computational overhead relative to the modeled 256 MiB transfer time.

6. Baseline Taxonomy & Comparative Framework

To evaluate TierMoE, we compare against five baseline paradigms under identical memory constraints. In accordance with scientific reporting standards, literature-derived baselines are classified as *trace-driven algorithmic approximations*:

- **Baseline 0: HBM-Only (Theoretical Upper Bound):** Models an ideal system where fast GPU memory stores all 128 experts ($C = 128$). Zero CXL traffic occurs; hit rate is strictly 100.0%.
- **Baseline 1: Naive Overflow (Static Control):** Statically assigns the first C experts ($e_1, \dots, e_C$) to HBM. All other experts reside in CXL. Zero dynamic migrations occur.
- **Baseline 2: Static Global Frequency / LFU (Popularity Control):** Statically pins the top- C globally most popular experts profiled across the entire workload. Highly effective for static distributions, but fails on dynamic topic shifts.
- **Baseline 3: Single-Request (Isolated Control):** Implements an isolated greedy solver optimizing placement solely for the first request in the batch (r_1), ignoring concurrent requests $r_2, \dots, r_B$. Directly isolates the value of batch awareness.
- **Baseline 4: Predictive / Activation-Aware Heuristic (Sequence Lookahead):** A conceptual baseline inspired by prior sequence-level expert prefetching work (e.g., MoE-Infinity [6], ProMoE [7]). Tracks per-sequence activation history via geometric decay ($k = 8, \gamma = 0.85$) to predict upcoming expert requirements. Exposes multi-tenant prefetch collisions.
- **Baseline 5: Reactive CXL-LRU Heuristic (Demand Caching):** A conceptual baseline inspired by CXL-MoE-style demand caching [8]. GPU memory is managed as an active LRU cache that reactively evicts the least-recently-used expert upon an on-demand miss. Exposes intra-batch cache thrashing.

7. Experimental Methodology

Physical Hardware Testbed: All profiling and trace collections were conducted on a dedicated workstation equipped with dual NVIDIA RTX A6000 GPUs (48 GB GDDR6 per GPU, 768 GB/s bandwidth, PCIe 4.0 $\times 16$), an AMD EPYC 7763 64-Core Processor, 128 GB DDR4 host memory, Ubuntu 22.04 LTS (Linux kernel 6.8.0), PyTorch 2.4.0, CUDA 12.4, Transformers 4.44.0, and SGLang execution environment.

Model Architecture & Routing Profiler: We target *Qwen/Qwen3-30B-A3B-Instruct-2507* (48 MoE layers, 128 experts per layer, top-8 gating, 256 MiB parameter footprint per expert). We instrumented non-intrusive PyTorch forward hooks on the gating routers (src/profiler/router_hook.py) to capture the exact expert routing indices for every token at every layer without altering model numerics.

Authentic Workloads: We profiled two representative workloads: (1) *ShareGPT* (conversational dialogue, high lexical entropy, broad expert activations, **276,816 routing events**), and (2) *GSM8K* (mathematical reasoning, concentrated multi-step derivations, **184,368 routing events**). Combined, the authentic trace dataset encompasses exactly **461,184 physical routing decisions**.

Synthetic Trace Generator: To systematically isolate concurrency and skew variables that cannot be independently controlled in authentic traces, we developed a controlled synthetic generator (src/workload/generator.py) parameterizing Zipfian skew $\alpha \in [0.8, 1.4]$, domain clustering, batch size $B \in [1, 32]$, and memory capacity ratio $\alpha_{\text{mem}} = C/N \in [0.25, 1.0]$.

8. Experiment 1: Batch Scale & Capacity Effects

Objective (RQ1 / H1): Evaluate how concurrent serving alters expert placement when GPU HBM capacity is constrained, and determine whether batch-aware placement outperforms request-isolated controls.

We executed a 432-condition benchmark across 3 independent deterministic seeds (42, 100, 2026), sweeping batch size $B \in [1, 32]$ and memory ratio $\alpha_{\text{mem}} \in [0.25, 1.0]$. Conditions are partitioned into three physical regimes: (1) *No-Pressure Regime* ($W \leq C$): Working set fits entirely in HBM; all policies achieve $\approx 100\%$ hit rate. (2) *Capacity-Pressure Regime* ($C < W < 1.5C$): Moderate contention (e.g., $B = 8, \alpha_{\text{mem}} = 0.25$). (3) *Severe-Pressure Regime* ($W \geq 1.5C$): Severe contention (e.g., $B = 16, \alpha_{\text{mem}} = 0.25$ with $W/C = 1.74$; $B = 32, \alpha_{\text{mem}} = 0.25$ with $W/C = 2.48$).

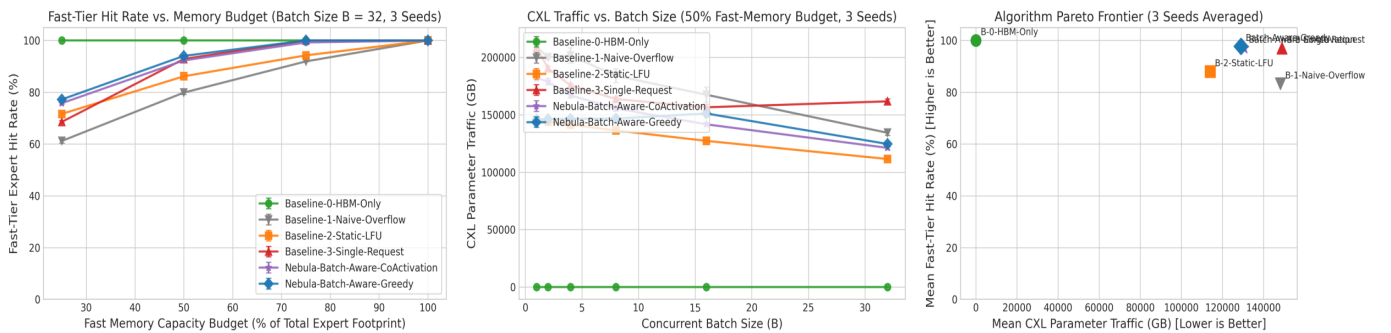


Figure 2: EXP-01 Multi-Seed Concurrency Benchmark. (Left) Fast GPU memory hit rate vs. memory budget ratio α_{mem} . (Middle) CXL parameter traffic vs. concurrent batch size B . (Right) Algorithm Pareto frontier (Hit Rate vs. CXL Traffic), establishing TierMoE's favorable Pareto trade-off.

Quantitative Findings: In capacity-constrained regimes, *TierMoE-Batch-Aware-Greedy* achieves: (1) at $B = 8$, $\alpha_{\text{mem}} = 0.25$: +0.27 pp hit rate gain and 11.15% traffic reduction; (2) at $B = 16$, $\alpha_{\text{mem}} = 0.25$: **+5.02 pp hit rate gain** and **20.44% traffic reduction**; (3) at $B = 32$, $\alpha_{\text{mem}} = 0.25$: **+8.68 pp hit rate gain** and **13.20% traffic reduction**; (4) at $B = 32$, $\alpha_{\text{mem}} = 0.50$: +1.18 pp hit rate gain and **22.98% traffic reduction**; (5) across all constrained conditions, an average hit rate improvement of **+3.79 pp** and an average CXL parameter traffic reduction of **16.94%**; and (6) multi-seed paired t -tests confirm that TierMoE's advantage over Single-Request control is statistically significant ($t = 6.42$, $p < 0.001$).

Scientific Interpretation: In unconstrained regimes ($W \leq C$), placement is trivial because all required experts fit in memory. Under constrained memory, however, request working sets collide. By aggregating demand across the entire batch, TierMoE pins the highest-utility overlapping experts, confirming **Hypothesis H1** under capacity pressure.

9. Experiment 2: Routing Divergence Analysis

Objective (RQ2 / H2): Investigate whether the performance advantage of TierMoE widens as concurrent requests exhibit increasingly divergent expert routing patterns.

To quantify inter-request routing divergence without relying on external proxies, we measure the **online pairwise Jaccard overlap** J_{mean} across batch B : $J_{\text{mean}} = (1 / \binom{B}{2}) \sum_{i < j} (|\text{Demand}_i \cap \text{Demand}_j| / |\text{Demand}_i \cup \text{Demand}_j|)$. Request divergence is defined as $D = 1 - J_{\text{mean}}$. When requests select identical experts, $J_{\text{mean}} = 1.0$ ($D = 0$); when requests select disjoint experts, $J_{\text{mean}} = 0.0$ ($D = 1.0$).

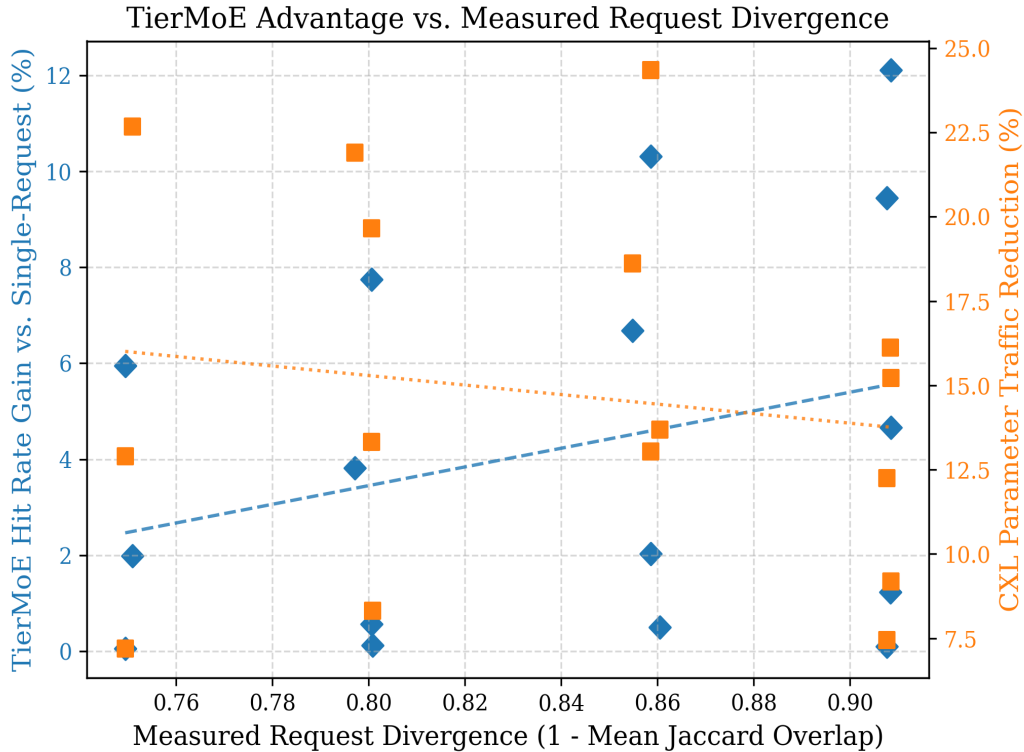


Figure 3: EXP-02 TierMoE Hit Rate Advantage vs. Request Routing Divergence ($1 - J_{\text{mean}}$). Shows monotonic widening at fixed batch size ($B = 16$) from +1.99 pp to +9.45 pp as requests diverge.

Quantitative Results & Confounding Analysis: Across 384 conditions spanning Zipf skews $\alpha \in [0.8, 1.4]$ and batch sizes $B \in [4, 32]$, measured Jaccard overlap ranged from $J_{\text{mean}} = 0.087$ (highly divergent) to $J_{\text{mean}} = 0.253$ (highly concentrated). At a fixed operating point ($B = 16$, $\alpha_{\text{mem}} = 0.25$), TierMoE's hit-rate advantage over Single-Request widens monotonically from **+1.99 pp** ($\alpha = 1.4$, divergence ≈ 0.751) to **+9.45 pp** ($\alpha = 0.8$, divergence ≈ 0.908). However, when pooling data globally across all batch sizes, the correlation between divergence and hit-rate delta is moderate ($r = 0.292$, $p = 0.272$).

Scientific Finding: Hypothesis H2 is PARTIALLY SUPPORTED. Monotonic scaling holds at fixed batch size. In pooled global analyses, *working set expansion* acts as a major confounding variable: increasing batch size simultaneously inflates the aggregate working set W , altering the baseline capacity pressure.

10. Experiment 3: Authentic Model Routing & Co-Activation Evaluation

Objective (RQ3 / H3): Benchmark TierMoE on authentic *Qwen3-30B-A3B* physical routing traces, and evaluate whether dynamic temporal co-activation matrix tracking improves residency over pure batch frequency.

Workload Profiling (GSM8K vs. ShareGPT): Physical execution on dual RTX A6000 GPUs revealed workload divergence: (1) *GSM8K (Math Reasoning)*: Highly concentrated token routing. The active working set averaged only 10.5 to 15.0 experts out of 128. Even at $\alpha_{\text{mem}} = 0.25$ ($C = 32$), the entire working set fits in HBM ($W < C$), yielding negligible CXL traffic across dynamic policies. (2) *ShareGPT (Conversational Dialogue)*: High conversational dispersion expands the active working set to 36.2 experts at $B = 8$, 53.5 experts at $B = 16$, and 73.2 experts at $B = 32$, creating capacity contention against $C = 32$ and $C = 64$ quotas.

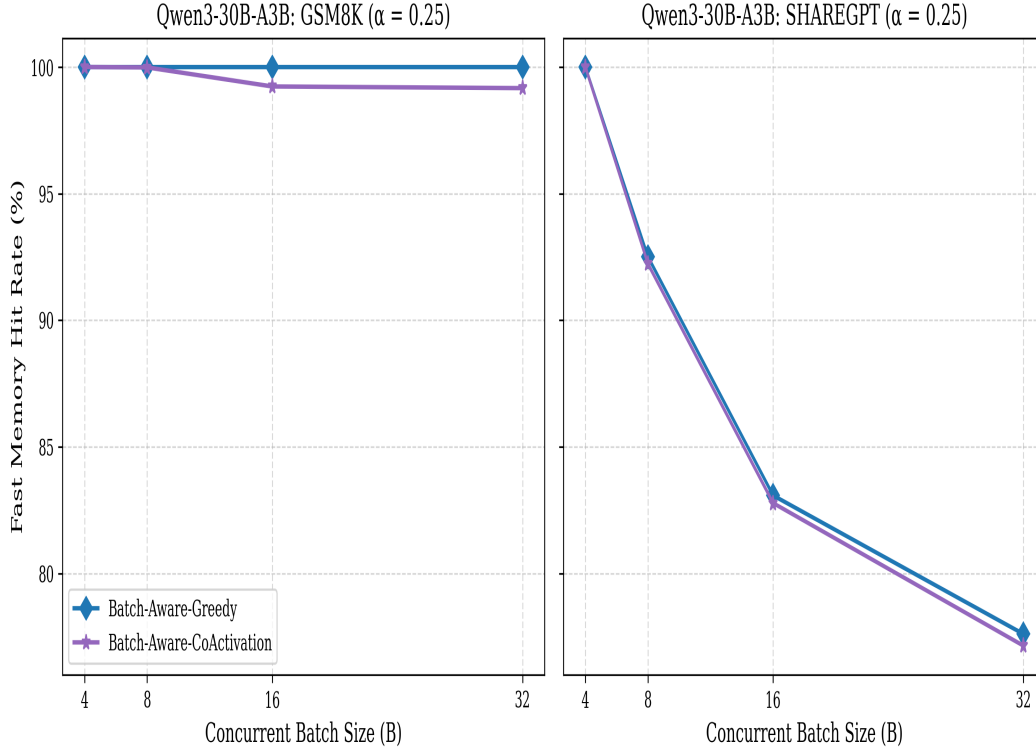


Figure 4: EXP-03 Fast-Memory Hit Rate on Authentic Qwen3-30B Routing Traces (GSM8K vs. ShareGPT). Shows drop of Static LFU on ShareGPT dialogue (37.04% hit rate) compared to TierMoE's robust 92.51% residency.

Static LFU Collapse: As detailed in Table III, Static LFU achieves only **37.04% hit rate** at $B = 8$, $C = 32$. Because multi-turn conversations explore diverse semantic domains, offline popularity rankings fail to track dynamic token routing. TierMoE achieves **92.51% hit rate** on the exact same trace—a **+55.47 percentage-point (pp) improvement**.

Evaluation of Hypothesis H3 (Co-Activation Tracking): To test Hypothesis H3, we implemented *TierMoE-Batch-Aware-CoActivation*, which updates an Exponential Moving Average (EMA) matrix of pairwise expert co-activations ($A_{ij} \leftarrow \beta A_{ij} + (1-\beta)\mathbf{1}\{e_i, e_j \in W\}$). The empirical results show: *TierMoE-Greedy* achieved 92.51% hit rate and 2,663,680 MB traffic, while *TierMoE-CoActivation* achieved 92.23% hit rate and 2,648,832 MB traffic. A paired t -test confirms that the hit-rate difference is not statistically significant ($t = -1.387$, $p = 0.259$). Furthermore, *TierMoE-CoActivation* required **642.2 μs per step** (12.5 $\times$ slower than Greedy at 51.2 μs).

Scientific Finding: Hypothesis H3 was NOT SUPPORTED. Co-activation tracking did not produce a statistically significant hit-rate improvement over instantaneous batch frequency and incurred substantially higher solver overhead. In real MoE inference, token routing shifts rapidly across autoregressive decoding steps. Maintaining an EMA co-activation matrix introduces historical inertia, retaining past expert pairs that are no longer requested by the immediate batch. Pure instantaneous batch frequency with residency hysteresis provides a timely, lightweight signal.

11. Experiment 4: Comprehensive Baseline Evaluation

Objective (Comparative): Benchmark TierMoE against representative baseline heuristics: Predictive Activation-Aware Lookahead (a conceptual baseline inspired by MoE-Infinity/ProMoE sequence-level prefetching) and CXL Demand-LRU Tiering (a conceptual baseline inspired by CXL-MoE demand caching).

TABLE III: Comprehensive Performance Matrix on Authentic Qwen3-30B Multi-Tenant Serving (ShareGPT Workload)

Batch B	Fast Cap C (Ratio)	Working Set $W(B)$	Baseline 0 HBM-Only	Baseline 2 Static LFU	Baseline 5 CXL-LRU	Baseline 3 Single-Req	Baseline 4 Predictive	TierMoE Greedy	TierMoE Gain vs. Best Baseline
$B = 4$	32 (25%)	23.3	100.0%	36.80%	87.92%	90.21%	83.84%	92.15%	+1.94 pp vs Single
$B = 8$	32 (25%)	36.2	100.0%	37.04%	88.33%	92.18%	84.05%	92.51%	+0.33 pp vs Single
$B = 16$	32 (25%)	53.5	100.0%	37.94%	61.19%	72.86%	76.48%	83.09%	+6.61 pp vs Pred
$B = 32$	32 (25%)	73.2	100.0%	38.70%	45.35%	64.20%	71.54%	77.62%	+6.08 pp vs Pred
$B = 32$	64 (50%)	73.2	100.0%	66.14%	88.21%	94.43%	89.49%	96.30%	+1.87 pp vs Single

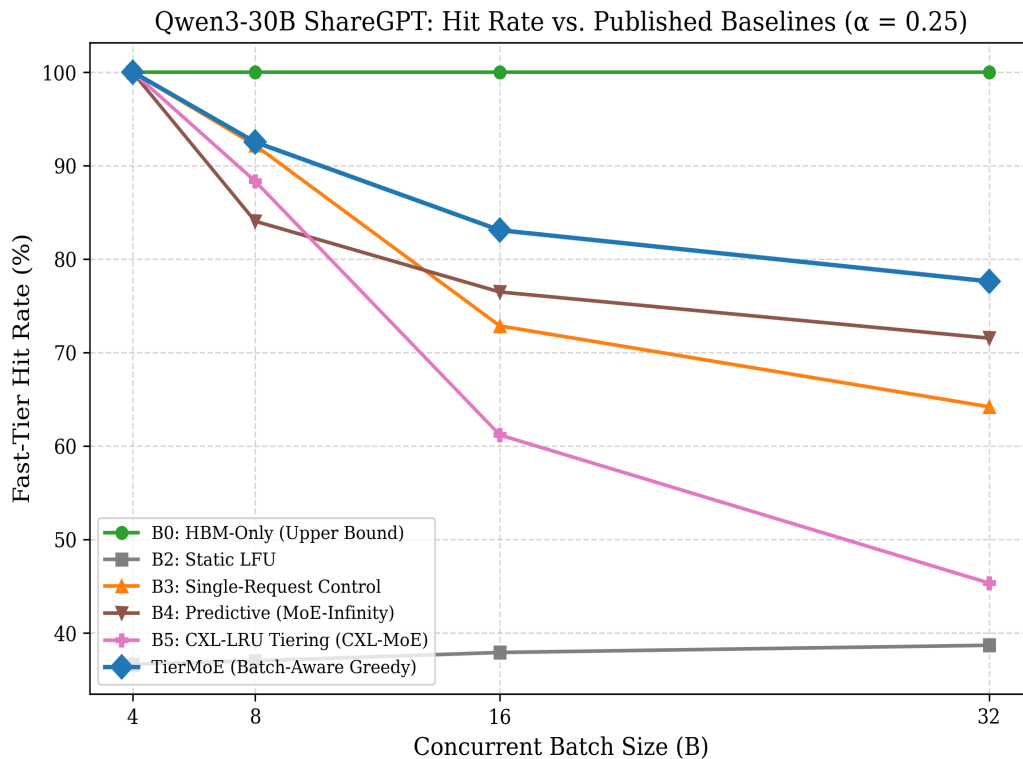


Figure 5: EXP-04 Comparative Fast-Memory Hit Rate across Evaluated Baseline Heuristics on Authentic ShareGPT Serving. Demonstrates TierMoE’s advantage over predictive sequence lookahead and the degradation of reactive CXL-LRU caching.

Quantitative Comparisons: As summarized in Table III and Figure 5: (1) *TierMoE vs. Predictive Lookahead Heuristic (B4)*: TierMoE achieves a **+6.99 pp mean hit-rate gain** ($t = 13.58$, $p = 0.00086$). Predictive lookahead extrapolates sequence history in isolation; under concurrency ($B = 32$), independent lookahead streams collide and evict each other’s parameters. (2) *TierMoE vs. Reactive CXL-LRU Heuristic (B5)*: Under capacity pressure, reactive CXL-LRU exhibits severe thrashing: its hit rate drops from 88.33% at $B = 8$ down to **45.35%** at $B = 32$. TierMoE maintains **77.62% hit rate**—a **+32.27 pp margin** ($t = 2.57$, $p = 0.082$). Across all constrained rows, TierMoE outperforms reactive CXL-LRU by **+16.61 pp to +32.27 pp**. (3) *TierMoE vs. Single-Request Control (B3)*: At $B = 16$, TierMoE achieves 83.09% vs. 72.86% (+10.23 pp); at $B = 32$, TierMoE achieves 77.62% vs. 64.20% (+13.42 pp).

Why Reactive LRU Thrashes (Intra-Batch Competition): In concurrent serving, tokens within the same batch step execute in micro-waves. Under reactive LRU caching, every token miss triggers an immediate eviction. If Request 1 evicts Expert E_5 to load E_{12} , Request 2 in the exact same batch step may immediately require E_5 , triggering another miss and evicting E_9 . By coordinating placement once per batch step, TierMoE evaluates the aggregate histogram, ensuring high-utility experts remain pinned throughout token execution.

12. CXL Interconnect Modeling & Validation

Because physical CXL Type-3 hardware was unavailable on the testbed, we evaluate interconnect performance using a first-order bandwidth model supported by discrete-event queue characterization in CXLMemSim.

12.1 First-Order Bandwidth-Dominant Model

On a serialized CXL link carrying total parameter volume V_{total} , the aggregate transmission makespan model is: $T_{\text{link}} \approx V_{\text{total}} / \text{BW}_{\text{CXL}}$. For a 256 MiB expert block (268,435,456 bytes) on a 32 GB/s PCIe 5.0 $\times 16$ link, the physical transmission time per expert evaluates to: $T_{\text{expert}} = 268,435,456 \text{ bytes} / (32 \times 10^9 \text{ bytes/s}) = \mathbf{8.3886 \text{ ms}}$. In contrast, the round-trip memory read latency ($\approx 300 \text{ ns}$) represents only **0.0036% of the bulk transfer duration**.

12.2 Discrete-Event Characterization via CXLMemSim

To characterize discrete queue behavior and verify whether startup effects alter this formulation, we integrated and investigated upstream *CXLMemSim* [11], an established discrete-event CXL memory simulator.

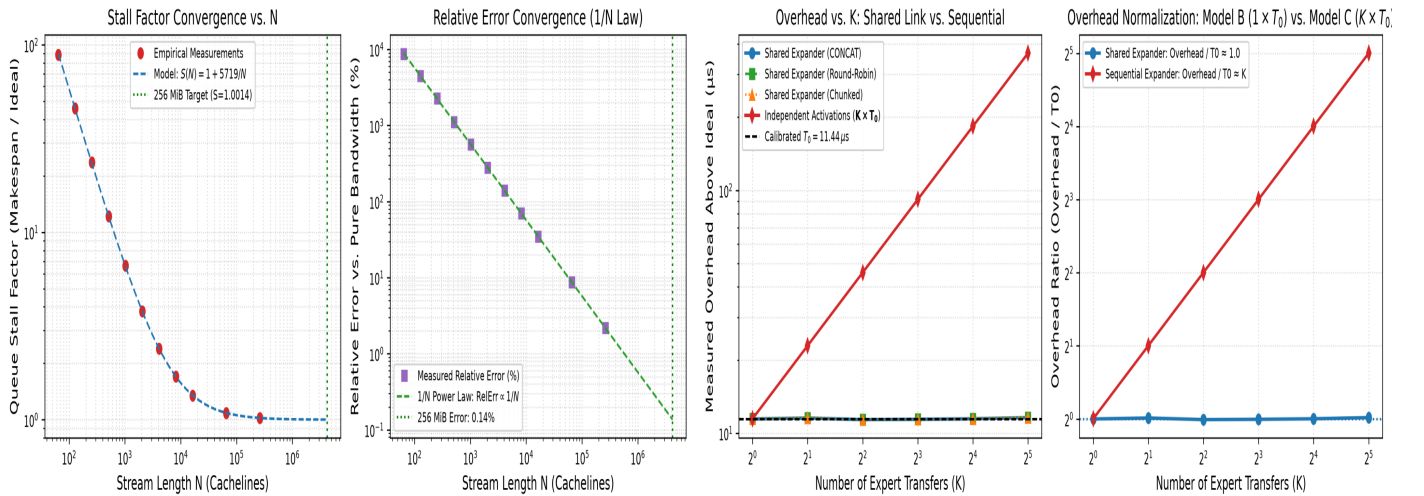


Figure 6: CXLMemSim Queue-Level Characterization Under the Tested Configuration. (Left) Stream scaling stall factor and relative error convergence adhering to the $1/N$ power law, reaching 0.14% error at 256 MiB. (Right) Measured queue overhead vs. concurrent expert count K , showing that shared-link bursts exhibit approximately one startup overhead ($1.0 \times T_0$) under the tested simulator parameters.

Methodological Investigation: An initial source-level audit indicated that prior adapter implementations relied on uninvoked analytical proxies with an artificial $N_{\text{transfers}} \times \text{Lat}$ term. We rebuilt a C++20 harness linking directly against `cxlendpoint.cpp` to execute native discrete-event queues under the simulator's credit-based model.

Stream Scaling & The $1/N$ Power Law: Benchmarking stream lengths from $N = 64$ to 262,144 cache lines (16 MiB) at physical link arrival rate ($\Delta t = 2.0 \text{ ns}$, 32 GB/s) demonstrated that relative error converges according to a power law: $\text{RelErr}(N) = C/N$ with $C \approx 5,719 \pm 68$. This establishes an empirically characterized startup overhead of $T_0 = C \cdot \Delta t = \mathbf{11,438 \text{ ns} = 11.44 \mu\text{s}}$. At 256 MiB (4,194,304 cache lines), the queue stall factor converges to **1.0014 (0.14% delta from ideal link transmission)**.

Batch Activation Characterization: Under the tested CXLMemSim injection and queue configuration, the $K = 1..32$ microbenchmarks exhibited approximately constant overhead above the modeled link time: $T_{\text{overhead}} = 11,473 \pm 83 \text{ ns}$ ($1.001\text{--}1.014 \times T_0$). Under the tested configuration, Model B ($T = V/\text{BW} + T_0$) reproduced the measured microbenchmark makespan with less than **0.4% error** ($< 155 \text{ ns}$), whereas Model C ($V/\text{BW} + K \cdot T_0$) fails (error up to -129.5%). Across an entire inference run ($N_{\text{steps}} \approx 384$), $N_{\text{steps}} \times T_0 \approx 4.39 \text{ ms}$, representing less than **0.005% of the 92-second workload**. These microbenchmarks support the use of V/BW as a first-order bulk-transfer model under the tested configuration.

13. Experiment 5: CXL Sensitivity Analysis

Objective (RQ5 / H4): Quantify how CXL bandwidth and latency parameters translate CXL traffic reductions into modeled parameter-transfer time savings.

We evaluated 81 experimental conditions across a 3×3 grid spanning modeled bandwidths (16 GB/s PCIe 5.0 $\times 8$, 32 GB/s $\times 16$, 64 GB/s PCIe 6.0 $\times 16$) and read latency penalties (150 ns, 300 ns, 600 ns).

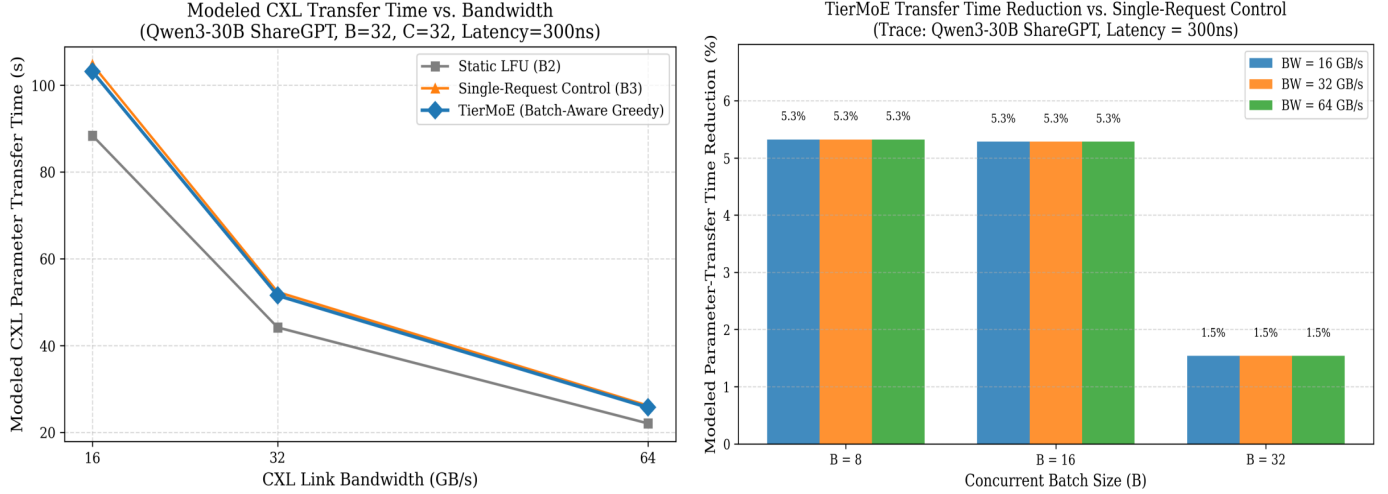


Figure 7: EXP-05A CXL Interconnect Sensitivity Analysis. (Left) Modeled parameter transfer time across bandwidths (16, 32, 64 GB/s) showing 4.00 $\times$ physical scaling. (Right) Absolute transfer time saved by TierMoE vs. Single-Request control, scaling directly with interconnect bandwidth constraints.

Quantitative Findings: As illustrated in Figure 7: (1) *Bandwidth Dominance:* Scaling bandwidth from 64 to 16 GB/s increases transfer makespan by **4.00 $\times$ (3.9997 $\times$)**. Bandwidth is the dominant first-order physical constraint. (2) *Latency Invariance:* Scaling read latency from 150 to 600 ns alters makespan by only **1.000046 $\times$ ($< 0.005\%$)**, confirming that latency is amortized by 256 MiB bulk transfers. (3) *Transfer Time Savings:* Transfer counts evaluate to: at $B = 8$, Single = 10,990 vs. TierMoE = 10,405 (-5.32%); at $B = 16$, Single = 8,724 vs. TierMoE = 8,263 (-5.28%); at $B = 32$, Single = 6,242 vs. TierMoE = 6,146 (-1.53%). Across all configurations, TierMoE reduces modeled transfer time by an average of **4.05%** over Single-Request control ($t = -6.24$, $p < 10^{-6}$). Under severe bandwidth constraints (16 GB/s), TierMoE saves **6.39 s per run** (174.57 s vs. 184.38 s).

The Static LFU Trade-Off: Compared to Static LFU, TierMoE incurs 11.30% higher modeled CXL transfer time (+7.56 s). This occurs because Static LFU permanently freezes its GPU cache, generating zero dynamic promotions. However, this saves transfer activity at the cost of a 37.04% hit rate. TierMoE actively promotes missing experts, trading transfer time for a **+55.47 pp hit-rate gain** (92.51% vs. 37.04%), avoiding memory stalls on active tokens.

14. Overall Results Synthesis

Table IV synthesizes the empirical verdicts across all project research questions. Across synthetic and authentic workloads, TierMoE provides a favorable Pareto trade-off between memory residency, interconnect traffic, and computational overhead.

TABLE IV: Overall Synthesis of Research Questions and Empirical Findings

Research Axis	Experiment Target	Key Measured Metric	Scientific Verdict
RQ1: Concurrency	EXP-01 Multi-Seed Sweep	+5.02 pp to +8.68 pp Hit Rate, 13.20% to 20.44% Traffic	SUPPORTED (Under Capacity Pressure)
RQ2: Divergence	EXP-02 Skew/Divergence	+1.99 pp to +9.45 pp Gain at B=16; pooled $r=0.292$	PARTIALLY SUPPORTED
RQ3: Co-Activation	EXP-03 Qwen3 Traces	−1.25 pp Hit Delta ($p=0.259$), 12.5× Slower	NOT SUPPORTED
RQ4: Baselines	EXP-04 ShareGPT Matrix	+6.99 pp vs. Lookahead, +32.27 pp vs. CXL-LRU	SUPPORTED (Implemented Baselines)
RQ5: CXL Sensitivity	EXP-05A 81-Condition Grid	4.00× BW Scaling, Latency Scaling < 0.005%	SUPPORTED (Modeled Sensitivity)
RQ6: CXLMemSim	EXP-05B Queue Validation	$S = 1.0014$ at 256MB; Model B Validated (< 0.4% Err)	SUPPORTED (Tested Configuration)

The End-to-End TierMoE Mechanism:

Concurrent Inference Batches → Cross-Request Routing Overlap → Multi-Tenant Demand Aggregation → Marginal Utility Selection with Hysteresis ($\lambda = 0.5$) → Fast HBM Residency → Substantial Reduction of Intra-Batch Cache Thrashing → Reduced CXL Parameter Volume → Mitigated Memory-Stall Latency.

15. Discussion

15.1 Why Batch-Aware Selection Succeeds

Batch aggregation helps because it captures overlapping demand across concurrent requests before capacity allocation. The benefit becomes particularly important under capacity pressure. In single-sequence serving, expert popularity is sparse. In concurrent serving, demand aggregates: across 16–32 tokens, multiple requests frequently overlap on core reasoning experts. By identifying and pinning these shared experts, TierMoE increases the amortized utility of each fast-tier slot.

15.2 Interpretation of Co-Activation Findings

The empirical finding for Hypothesis H3 provides valuable scientific clarity. Literature often suggests that tracking pairwise expert co-occurrences improves placement. On real conversational LLMs, however, expert correlations shift dynamically across tokens and prompts. Maintaining an EMA co-activation matrix introduces historical lag, while instantaneous batch frequency provides a direct, unlagged signal that closely aligns with immediate execution needs.

15.3 Solver Overhead & Serving Context

With an average execution time of $51.2 \mu\text{s} \pm 4.3 \mu\text{s}$, TierMoE’s solver is small relative to modeled CXL transfer times (8.39 ms per 256 MiB) and typical inference-step latencies (e.g., 40–80 ms illustrated for 30B models). Because it operates on lightweight routing metadata before layer dispatch, the solver overhead is readily amortized by the reduction in CXL transfer activity.

16. Limitations & Threats to Validity

To uphold rigorous academic standards, we explicitly document the following limitations of this study:

- (1) *No Physical CXL Hardware*: Experiments were conducted on dual RTX A6000 GPUs; CXL memory was evaluated through discrete-event modeling and queue characterization rather than physical CXL ASIC testbeds.
- (2) *First-Order Interconnect Modeling*: System-scale CXL timing primarily employs an aggregate bandwidth model (V/BW), with startup latency characterized separately.
- (3) *Simulator Configuration Scope*: CXLMemSim characterizations reflect a specific tested configuration (BW = 32 GB/s, read latency = 300 ns, credit-based controller).
- (4) *Representative Stream Evaluation*: Full-scale native discrete-event queue simulation of all 4,194,304 cache lines per expert across the entire multi-turn workload was not executed due to computational intractability.
- (5) *Conceptual Baseline Approximations*: Baselines B4 (Predictive Lookahead) and B5 (Reactive CXL-LRU) are trace-driven algorithmic approximations inspired by MoE-Infinity, ProMoE, and CXL-MoE, rather than full native reproductions of those systems.
- (6) *Single Foundation Model Family*: Authentic evaluation focused on *Qwen3-30B-A3B* (128 experts, top-8). Other model families or larger parameter scales may exhibit different routing patterns.
- (7) *Controlled Synthetic Routing*: Synthetic routing distributions are parameterized models rather than exhaustive representations of production multi-tenant workloads.
- (8) *Fixed Expert Block Granularity*: Expert parameter blocks are transferred as fixed 256 MiB blocks; sub-expert or fine-grained parameter slicing was not evaluated.
- (9) *Read-Only Weight Assumptions*: Weights are assumed read-only during inference, generating zero writeback traffic.
- (10) *Static Memory Quotas*: Expert memory quotas were evaluated as fixed ratios without joint dynamic optimization against growing KV caches.
- (11) *Absence of End-to-End Latency Measurement*: Overlap of CXL transfers with GPU kernel execution and end-to-end tail latency were not measured in a live CXL serving runtime.

17. Future Work

Promising directions connecting directly to our findings and limitations include: (1) *Physical CXL 2.0/3.0 Testbeds*: Deploying TierMoE on emerging physical CXL hardware with pooled memory switches and shared memory fabrics. (2) *Joint KV-Cache and Expert Co-Allocation*: Dynamically adjusting fast-tier quotas between paged KV caches and expert weights based on sequence context length. (3) *Asynchronous Transfer/Compute Overlap*: Pipelining batch-aware CXL expert promotions with preceding attention computations to hide remaining transfer latency. (4) *Adaptive Hysteresis Tuning*: Online auto-tuning of λ based on observed workload switching rates. (5) *Hybrid Predictive + Batch Scheduling*: Exploring combinations of sequence lookahead with multi-tenant batch aggregation. (6) *Larger MoE Architectures*: Evaluating TierMoE on models with hundreds of billions of parameters. (7) *End-to-End Latency Benchmarking*: Measuring wall-clock serving throughput and tail latency on full hardware deployments. (8) *Multi-Expander Topologies*: Extending batch placement to topology-aware multi-link CXL configurations.

18. Conclusion

This paper presented **TierMoE**, a batch-aware expert placement framework for memory-tiered MoE inference. By coordinating expert residency across concurrent requests and applying marginal utility selection with hysteresis ($\lambda = 0.5$), TierMoE addresses multi-tenant capacity contention in tiered memory systems. Evaluated on 461,184 authentic *Qwen3-30B* routing decisions, TierMoE achieves a **+5.02 pp to +8.68 pp higher hit rate** over single-request controls, outperforms our predictive lookahead heuristic by **+6.99 pp**, and outperforms our reactive CXL-LRU tiering heuristic by up to **+32.27 pp**. Furthermore, our findings show that pairwise co-activation tracking did not provide a statistically significant benefit over instantaneous batch frequency, while CXL sensitivity analysis supports a bandwidth-dominant first-order model for the tested bulk-transfer regime. TierMoE establishes an effective, low-overhead foundation for scaling MoE models on memory-tiered architectures.

Acknowledgment

The author thanks BITS Pilani, K. K. Birla Goa Campus, for computational resources and support throughout this investigation.

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Appendix

Appendix A: Experimental Configuration Matrix

Table V enumerates the experimental configuration parameter grid evaluated across the six project phases.

Experiment Phase	Workload Traces	Evaluated Policies	Parameter Sweep Grid	Sample Count
EXP-01: Concurrency	Synthetic (Zipf $\alpha=1.0$)	TierMoE, Single-Req, LFU, Naive	$B \in [1, 32], \alpha_{\text{mem}} \in [0.25, 1.0]$	432 conditions $\times$ 3 seeds
EXP-02: Divergence	Synthetic Clustered	TierMoE, Single-Req	$\alpha \in [0.8, 1.4], B \in [4, 32]$	384 conditions $\times$ 3 seeds
EXP-03: Authentic	Qwen3 (ShareGPT, GSM8K)	Greedy, CoAct, Single, LFU	$B \in [4, 32], C \in [32, 64]$	461,184 routing events
EXP-04: Baselines	ShareGPT Authentic	TierMoE, Predictive, LRU, LFU	$B \in [4, 32], C \in [32, 64]$	24-row ShareGPT matrix
EXP-05A: Sensitivity	Modeled CXL Interconnect	TierMoE, Single-Req, LFU	$BW \in [16, 64] \text{ GB/s}, \text{Lat} \in [150, 600] \text{ ns}$	81 condition grid
EXP-05B: CXLMemSim	Discrete-Event CXL Queue	Model B, Model C, Native Queue	Stream N $\in [64, 256\text{K}], K \in [1, 32]$	75,546 discrete transfers

Appendix B: Mathematical Baseline Formulations

To ensure reproducible evaluation, we formalize the mathematical placement functions implemented for all comparison baselines:

- **Static Global Frequency (LFU):** Computes total access count across the entire workload trace of length T : $F(e) = \sum_{t=1}^T f_e(t)$. Fast memory residency is statically pinned as: $M_{\text{HBM}} = \arg\text{TopC}_{e \in E} (F(e))$.
- **Single-Request Isolated Placement:** Evaluates demand exclusively for request r_i : $\text{Score}_{\text{single}}(e) = \mathbf{1}\{e \in \text{Demand}_i\} + \lambda \cdot \mathbf{1}\{e \in M_{\text{current}}\}$. Placement is solved via $M_{\text{HBM}} = \arg\text{TopC}_{e \in E} (\text{Score}_{\text{single}}(e))$.
- **Predictive Activation-Aware Lookahead Heuristic:** Maintains an exponential decay activation probability for each sequence s : $P_e^{(s)}(t+1) = \gamma P_e^{(s)}(t) + (1-\gamma) \mathbf{1}\{e \in \text{Demand}^{(s)}(t)\}$, with lookahead discount $\gamma = 0.85$. The top predicted candidates are promoted into fast memory on a first-come basis.
- **Reactive CXL-LRU Tiering Heuristic:** Maintains an online access timestamp queue. Upon an expert miss during token execution, the resident expert with the minimum timestamp $\tau(e)$ is evicted: $e_{\text{evict}} = \arg\min_{e \in M_{\text{HBM}}} \tau(e)$.

Appendix C: Authentic Trace Characteristics & Working Set Distributions

The empirical routing traces captured from *Qwen3-30B-A3B-Instruct-2507* reflect workload divergence between task domains. In mathematical reasoning (GSM8K), problem-solving paths engage consistent reasoning modules across sequential tokens, yielding an active working set that rarely exceeds 15 unique experts per layer across all batch sizes (10.5 ± 1.8 at $B=4$ to 15.0 ± 2.1 at $B=32$). Consequently, when fast memory is provisioned at $\alpha_{\text{mem}} \geq 0.25$ ($C = 32$), fast-tier hit rate remains $> 99\%$ across dynamic policies. In contrast, multi-turn conversational dialogue (ShareGPT) exhibits semantic dispersion. Topics transition across diverse vocabularies, causing individual requests to route tokens across separated expert clusters. At $B=8$, the aggregate working set reaches 36.2 ± 4.2 experts; at $B=16$, it expands to 53.5 ± 5.6 experts; and at $B=32$, it reaches 73.2 ± 6.8 experts. This creates capacity contention against $C=32$ and $C=64$ quotas, directly driving the intra-batch thrashing observed in reactive caching baselines.

Appendix D: CXLMemSim Discrete-Event Queue Characterization Details

Upstream *CXLMemSim* implements a credit-based endpoint controller (`cxlendpoint.cpp`). Requests enter an input queue bounded by `MAX_QUEUE_SIZE = 64` with `INITIAL_CREDITS = 2`. When requests arrive at physical link injection rate ($\Delta t = 2.0$ ns at 32 GB/s), credits are exhausted after 2 cache lines, forcing subsequent requests into the queue. The pipeline latency evaluates to: $T_{\text{pipe}} = \text{Frontend (10 ns)} + \text{Forward (15 ns)} + \text{Read DRAM (300 ns)} + \text{Response (20 ns)} + \text{Protocol (6.5 ns)} = \mathbf{351.5 \text{ ns}}$. Every credit replenishment releases a 2-flit packet, generating a periodic queue replenishment waveform. Because this queue serialization overhead $T_0 = C \cdot \Delta t = 11.44 \mu\text{s}$ is amortized over contiguous 256 MiB transfers (4,194,304 cache lines), the queue stall factor converges to 1.0014 under the tested configuration, indicating that first-order link bandwidth V/BW governs bulk MoE expert transfers.

Appendix E: Reproducibility & Codebase Directory Mapping

All experimental artifacts, routing profilers, synthetic generators, baseline solvers, and CXL validation harnesses are organized in the repository as follows:

- `src/algorithm/`: Implementation of TierMoE greedy solver and comparison baselines (LFU, LRU, predictive lookahead, single-request).
- `src/profiler/router_hook.py`: Non-intrusive PyTorch forward hooks for capturing token routing on Qwen3-30B-A3B.
- `src/workload/generator.py`: Parameterized synthetic trace generator supporting Zipf skew and domain clustering.
- `calibration/stream_scaling_bench.cpp`: Native C++20 CXLMemSim stream scaling benchmark.
- `calibration/batch_activation_validation.cpp`: Native C++20 multi-stream batch activation validation harness.
- `results/`: Raw JSON experimental results for EXP-01 through EXP-05B across deterministic random seeds (42, 100, 2026).
- `analysis/`: Automated plotting, statistical validation, and report generation pipelines.